Network Basics

Level 0–1 IT Basics - 04 Networking

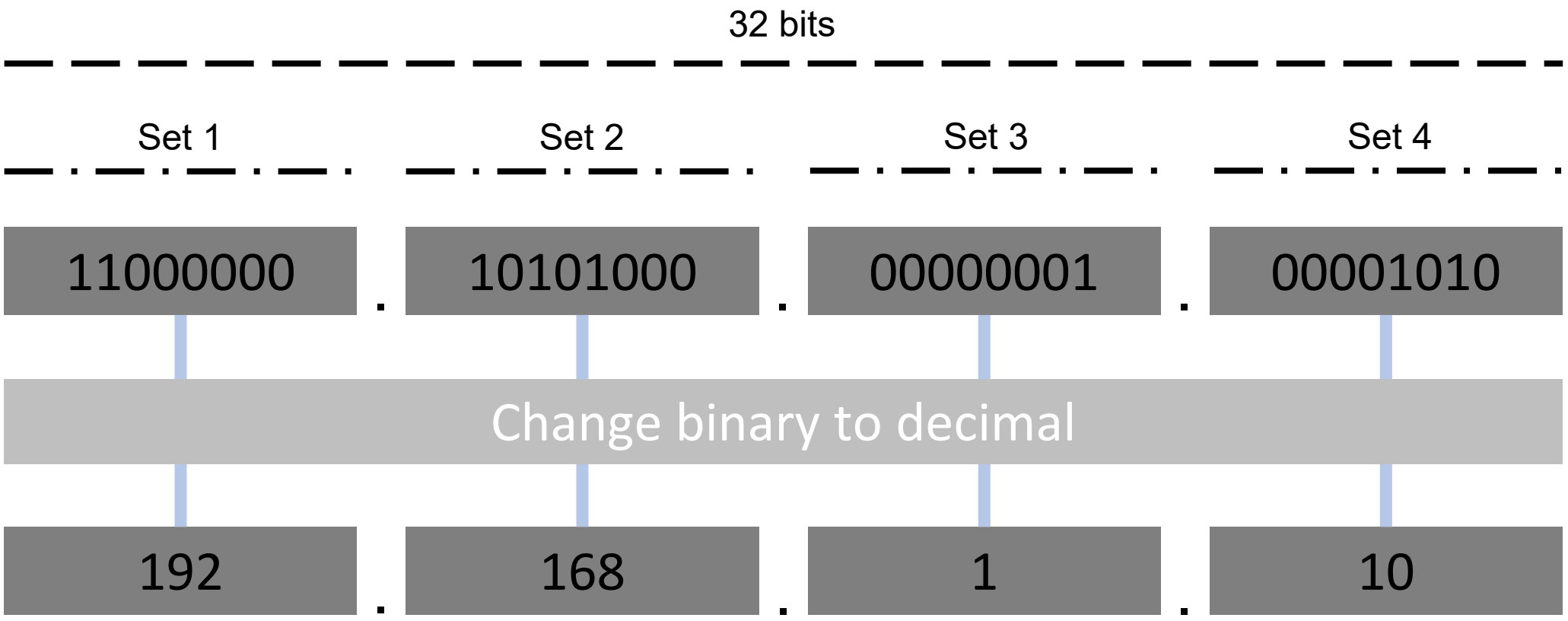
Security Level:

HUAWEI

What is Network?

IP Address format: IPv4

IPv4 has a total of 32 bits, divided into 4 sets, each containing 8 bits.



Networking Basics — Subnet Mask of an IP Address

- **A subnet mask**, also known as netmask or bitmask, is a 32-bit address, segregating an IP address into network bits that identify the network and host bits that identify the host. A subnet mask must be used together with an IP address.
- A **subnet mask** has 32 bits, much the same as that of an IP address.
- In binary mode, a **subnet mask** is represented by a group of consecutive 1s followed by a group of consecutive 0s.
- In a network mask, bits with the value of 1 correspond to the network part in an IP address, while bits with the value of 0 correspond to the host part. This helps identify the network and host parts in an IP address.

IP address	**192.168.1.1**	1 1 0 0 0 0 0 0	1 0 0 0 0 0 0 0	0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 1
Subnet mask	**255.255.255.0**	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	0 0 0 0 0 0 0 0
				Network part	**Host part**

By default,

- **Class A IP addresses**: The first octet represents the network ID and the remaining three octets indicate the host ID. The default mask is 255.0.0.0 or /8.
- **Class B IP addresses**: The first two octets represent the network ID and the remaining two octets indicate the host ID. The default mask is 255.255.0.0 or /16.
- **Class C IP addresses**: The first three octets represent the network ID and the last octet indicates the host ID. The default mask is 255.255.255.0 or /24.

If you apply for a class A address range 123.0.0.0/8, you will get 2^{24} IP addresses, 16,777,216 addresses to be exact. Relatively, a class B IP address range has 2^{16} IP addresses, and a class C IP address range has 2^{8} IP addresses.

Tip: 192.168.1.1 255.255.255.0 is equivalent to 192.168.1.1/24.

Subnet mask example

IP Network: 192.168.0.0/24 Subnet mask: 255.255.255.0

Network	Host	IP Address
192.168.0	00000000	192.168.0.0
192.168.0	00000001	192.168.0.1
192.168.0	00000010	192.168.0.2
….	….	….
192.168.0	1111110	192.168.0.253
192.168.0	1111101	192.168.0.254
192.168.0	1111111	192.168.0.255

Available IP 251

x.x.x.0: Network ID. This address is the beginning of the private IP address range and will not be assigned to any instance.

x.x.x.1: The gateway address of the subnet.

x.x.x.253: Reserved for the system interface. This IP address is used by the VPC for external communication.

x.x.x.254: DHCP service address.

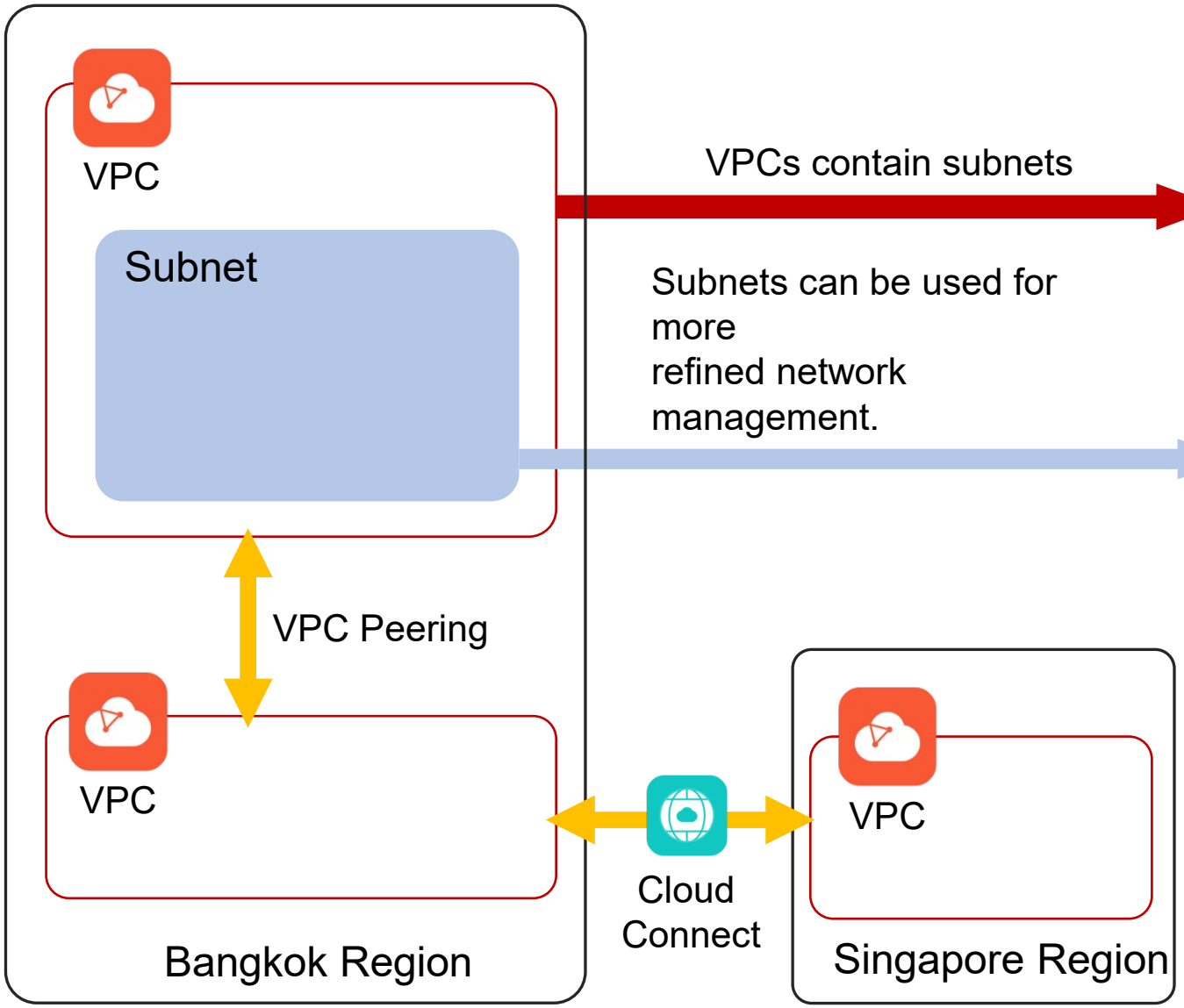
x.x.x.255: Network broadcast address.

Virtual Private Cloud

Virtual Private Cloud (VPC)

- "LANs" on the cloud
  - Resources in a VPC can communicate with each other.
  - Resources from **different VPCs** are **isolated** from each other.
- Custom IP address allocation and route configuration
- Core network security service
- If two VPCs are in the **same region**, use a **VPC peering** connection.
- If two VPCs are in **different regions**, use **Cloud Connect.**

Relationships Between VPCs and Subnets



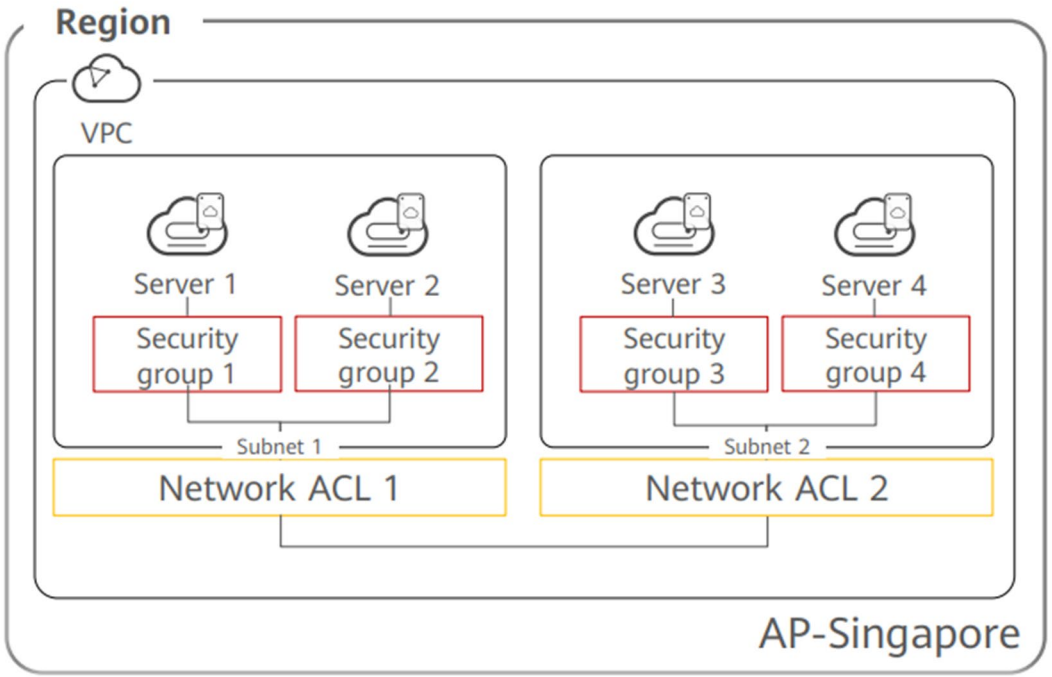
VPCs contain subnets

- You specify private IP address ranges for your VPCs.
- Resources in a VPC can communicate with each other.

Subnets can be used for more refined network management.

- A subnet is a subset of its VPC.
- You can deploy resources of the same type in the same subnet for easier management.
- Traffic can be controlled in and out of subnets.
- You can customize routes for different network scenarios
- All resources must be created in subnets

Security Control in a VPC



- Network ACLs and security groups are similar to firewalls.
- Security groups protect instances and check traffic both to and from instances.
- Network ACLs protect subnets and only check traffic across subnets.
- Security groups can be configured to allow access from
IP address ranges or other security groups.

Elastic IP & NAT Gateway

Elastic IP
(EIP)

- **Dynamic BGP**
- **Static public IP addresses**
- Scalable bandwidths
- If a resource has an EIP bound, it can directly **access the Internet.**
- EIPs can be bound to or unbound from ECSs, BMSs, virtual IP addresses, NAT gateways, or load balancers.

NAT Gateway

- A network address translation (NAT) service
- A public NAT gateway enables cloud and on-premises servers in **a private subnet to share an EIP to access the Internet** or **provide services accessible from the Internet**.
- A Private NAT gateways provide network address translation, allowing ECSs and BMSs in a VPC to communicate with servers in other VPCs or on-premises data centers.
- **SNAT** translates **private IP addresses into EIPs**
- **DNAT** enables servers within an AZ or across AZs in a VPC to share an EIP to **provide services accessible from the Internet**.

Direct Connect & Virtual Private Network

Direct Connect

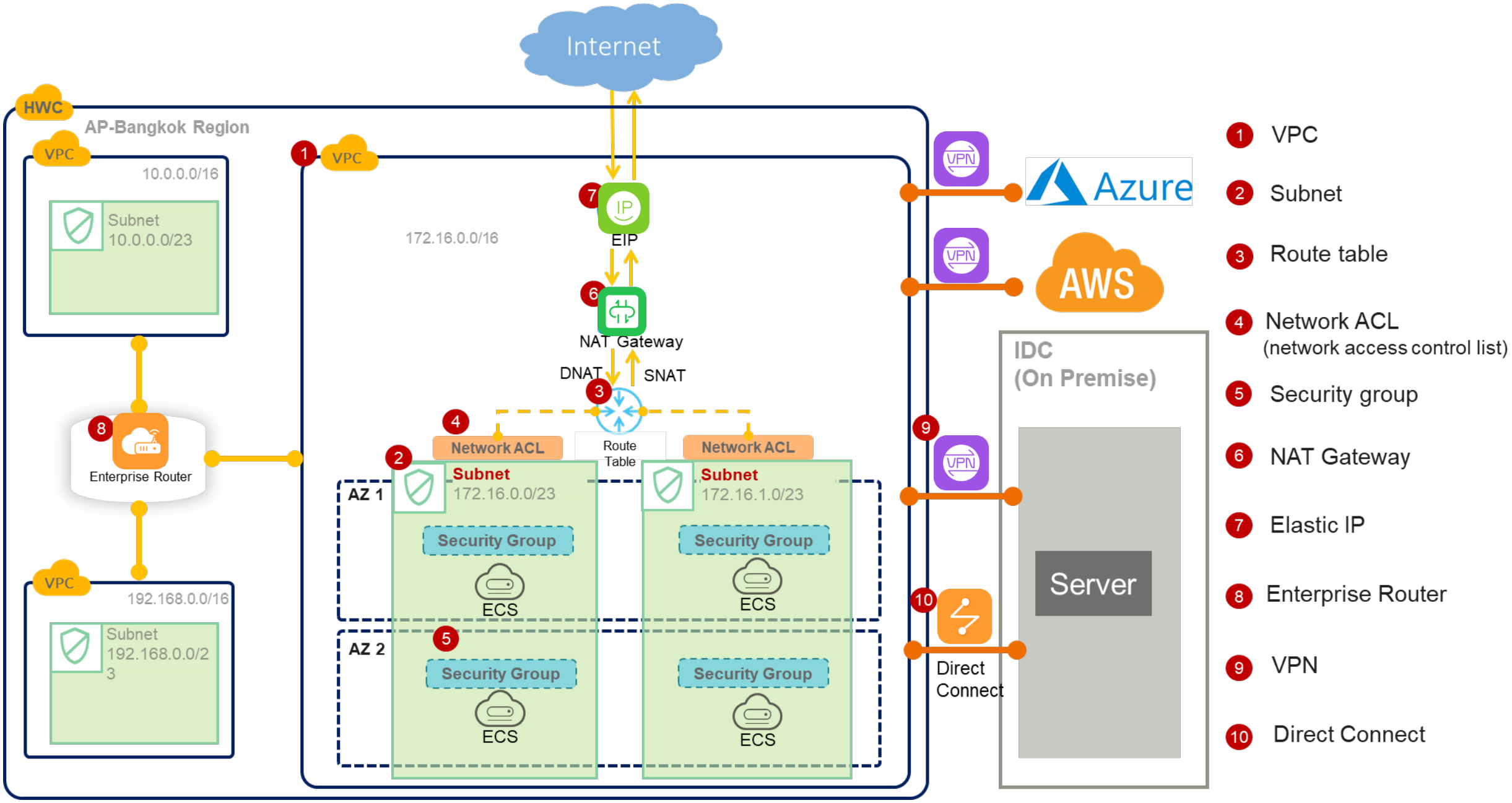
- A dedicated network is used for data transmission, which brings high network performance, low latency, and excellent user experience
- Direct Connect establishes private connectivity between an on-premises data center and the cloud. Data is transmitted over a dedicated connection, meeting the requirements for compliance.
- A single connection supports up to 100Gbit/s of bandwidth, which meets a diverse range of bandwidth requirements

Virtual Private Network (VPN)

- secure, reliable, and cost-effective encrypted connections between your on-premises network or data center and a virtual network on Huawei Cloud.

Enterprise Router (ER)

- An enterprise router connects virtual private clouds (VPCs) and on-premises networks to build a central hub network.


Internet
HWC
AP-Bangkok Region
VPC
10.0.0.0/16
Subnet
10.0.0.0/23
1
VPC
172.16.0.0/16
7
EIP
6
NAT Gateway
DNAT
SNAT
3
4
Network ACL
Route Table
Network ACL
2
Subnet
172.16.0.0/23
Subnet
172.16.1.0/23
AZ 1
Security Group
ECS
Security Group
ECS
AZ 2
5
Security Group
ECS
Security Group
ECS
8
Enterprise Router
VPC
192.168.0.0/16
Subnet
192.168.0.0/23
VPN
Azure
VPN
AWS
9
VPN
IDC
(On Premise)
Server
10
Direct Connect
1 VPC
2 Subnet
3 Route table
4 Network ACL
(network access control list)
5 Security group
6 NAT Gateway
7 Elastic IP
8 Enterprise Router
9 VPN
10 Direct Connect